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HKRG: Hierarchical knowledge integration for radiology report generation

Bo Wang [a,d], Peihong Teng [e], Hongda Zhang [c,d], Feiyang Yang [b], Zeyu Wang [g], Xingcheng Yi [f], Tianyang Zhang [i], Chunhui Wang [h], Adriano Jose Tavares [a], Hao Xu [b,d,*]

[a] School of Engineering, University of Minho, Braga, 4710-057, Portugal
[b] College of Computer Science and Technology, Jilin University, Changchun, 130012, Jilin, China
[c] School of Artificial Intelligence, Jilin University, Changchun, 130012, Jilin, China
[d] Joint International Research Laboratory of Ageing Active Strategy and Bionic Health in Northeast Asia of Ministry of Education, Jilin University, Changchun, 130012, Jilin, China
[e] Department of Radiology, China-Japan Union Hospital of Jilin University, Changchun, 130033, Jilin, China
[f] Laboratory of Cancer Precision Medicine, the First Hospital of Jilin University, Changchun, 130013, Jilin, China
[g] College of Computer Science and Engineering, Dalian Minzu University, Dalian, 116600, Liaoning, China
[h] School of Software, Jilin University, Changchun, 130012, Jilin, China
[i] Department of General Surgery Center Hepatobiliary and Pancreatic Surgery, The First Hospital of Jilin University, Changchun, 130021, Jilin, China

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ABSTRACT

Radiology report generation aims to automatically produce coherent, free-text reports that describe the clinical observations of radiographs. Most existing approaches align visual features with full report features to bridge the gap between medical images and reports. However, while the translation from concrete medical images to abstract textual reports requires recognizing anatomical structures, identifying lesions by their morphology and location, and interpreting their clinical significance, models often struggle to achieve precise anatomical alignment. This limitation results in overly generalized reports that lack clinical detail. To tackle this, we propose the hierarchical knowledge integration framework (HKRG), which facilitates the generation of radiology reports by guiding the process through layered domain knowledge. We extract the correspondence between organs and pathologies, serving as a buffer for the conversion between image–text pairs. Additionally, a cross-modal fusion module and image–text matching tasks embed this knowledge into the report generation process, learning implicit connections between images and clinical information. Experimental results on IU-Xray and MIMIC-CXR datasets show that our method outperforms previous state-of-the-art models, with notable improvements in key performance metrics. Ablation studies confirm the effectiveness of our framework, with an 11%–17% improvement in BLEU-4 and ROUGE-L scores compared to baseline models. By enhancing the accuracy of radiology report generation, our framework has the potential to reduce diagnostic errors and improve clinical decision-making. Codes are available at https://github.com/LuckyAI-wb/HKRG.

1. Introduction

Radiology imaging is a widely adopted non-invasive diagnostic tool in clinical settings. In recent years, the automatic generation of radiology reports has gained significant value in the medical field, primarily to enhance diagnostic efficiency, improve report consistency and standardization, and minimize human errors (Jain et al., 2021; Sloan, Clatworthy, Simpson, & Mirmehdi, 2024; Wang & Summers, 2012). However, due to the complexity (i.e., radiological images are highly complex and contain subtle nuances) and specificity (i.e., no two patients or cases are exactly alike) of the data involved (Delrue et al., 2011; Yang et al., 2023), it is difficult for automated systems to provide accurate and reliable reports in different situations.

Existing methods for automatically generating radiology reports primarily rely on multi-modal learning of visual and language features. This approach comprehensively analyzes the visual characteristics of images along with corresponding linguistic descriptions to achieve a deep understanding of radiological images. For example, methods like CDGPT2 (Alfarghaly, Khaled, Elkorany, Helal, & Fahmy, 2021), R2Gen (Chen, Song, Chang, & Wan, 2020), and R2GenCMN (Chen, Shen, Song, & Wan, 2022b) enhance multi-modal fusion through improved end-to-end structures and memory networks, thereby improving the quality of report generation. Simultaneously, studies (Huang, Zhang, & Zhang, 2023; Liu, Wu, Ge, Fan, & Zou, 2021b; Zhan, Lin,

* Corresponding author.
E-mail address: xuhao@jlu.edu.cn (H. Xu).

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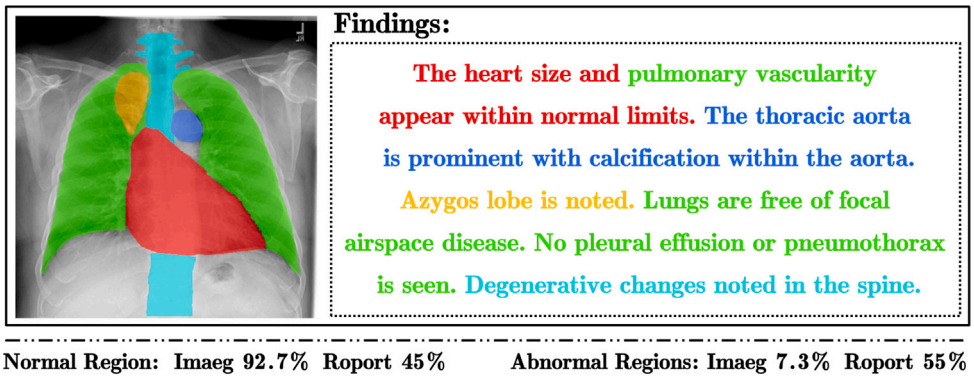


Fig. 1. Illustration of the areas in radiology reports and images, where the normal areas (red, green) occupy a larger proportion in the images, yet the descriptions of abnormal areas (yellow, dark blue, light blue) are more frequent in the text reports.

Wang, Wang, & Wu, 2024) utilize disease labels and cross-modal learning to enhance the system's ability to process and integrate various clinical information. However, establishing connections between images and reports often overlooks the pixel-level characteristics of anatomical structures, leading to two problems: (1) Image features often contain subtle visual cues that can indicate important clinical information. Critical diagnostic details might be omitted if these cues are not effectively captured and translated into the textual report. This can result in generic or incomplete reports that lack the specificity required for precise medical interpretation. (2) Images may require interpretation that relies on spatial relationships and the presence or absence of certain features, which can be difficult to translate into text accurately. Without a deep integration of visual and textual understanding, the generated reports might miss or misrepresent the clinical significance of these features. As shown in Fig. 1.

To tackle this, we reconsider the data bias between medical images and reports. Our strategy involves a two-stage joint training of data from different modalities, enabling heterogeneous data to establish connections within the feature space (i.e., initially aligning images with pathological anatomy prior knowledge, followed by aligning the images with text). We propose a Hierarchical Knowledge Integration Framework (HKRG), which utilizes layered domain knowledge to guide the process of generating radiology reports, thereby enabling the model to more accurately identify critical diagnostic features and reduce information gaps. Specifically, our method is divided into two main stages: image–text pre-training and radiology report generation. During the image–text pre-training stage, multiple objectives are optimized to align images, knowledge, and text at different granularity levels, ensuring effective integration of visual and language features. In the radiology report generation stage, we integrate hierarchical knowledge with a cross-modal fusion module, embedding these correspondences as model inputs into the report generation process to explore the implicit connections between images and knowledge.

Our contributions can be summarized as follows:

- We propose the hierarchical knowledge integration Framework (HKRG), which guides the generation of radiology reports through layered domain knowledge. This framework integrates multi-level knowledge and visual information to incorporate semantic features into images, effectively mitigating the disparity between image perception and text cognition, and strengthening the consistency between image and text.
- We proposed a multi-task learning-based method for image–text pre-training. This method achieves multi-level alignment by constructing tasks that align images with domain knowledge and textual reports. We embed this knowledge into the report generation process as model inputs, facilitating the learning of implicit connections between medical images and medical priors.
- Experiments and analyses on the public IU-Xray and MIMIC-CXR datasets have validated the effectiveness of the HKRG framework, demonstrating superior performance compared to previous state-of-the-art models on both datasets.

In Section 2, we conducted an in-depth discussion on prior work in the field of radiology report generation. In Section 3, we provide a detailed introduction to the proposed model, including its workflow, various modules, and technical details. The experimental results and ablation studies are presented in Section 4. Finally, the conclusion and future work are outlined in Section 5.

2. Related work

This section reviews recent work on radiology report generation and vision-language pre-training. We focus on these two aspects as they are crucial for understanding the advancements and challenges in integrating visual and textual data for radiology report generation.

2.1. Radiology report generation

In the field of radiology report generation, recent advancements in deep learning have predominantly focused on two principal directions: (1) enhancement model architectures, and (2) integrating external modal information. These methods aim to enhance feature fusion between different modalities or to leverage rich external data sources to augment the quality and accuracy of generated reports.

Several studies have introduced innovative architectures that strengthen the fusion of image and textual features:

- **CDGPT2** (Alfarghaly et al., 2021): Enhances feature integration by concatenating visual and semantic features within a multi-modal decoder, improving the model's capability to synthesize information.
- **AlignTransformer** (You et al., 2021): Utilizes a Multi-Grained Transformer to fuse features at different levels, enhancing the handling of complex medical images and associated texts.
- **R2Gen** (Chen et al., 2020) & **R2GenCMN** (Chen et al., 2022b): Combines specialized LSTM and CMN modules to process image and text separately, enhancing the relevance and accuracy of generated reports by refining extracted features.

In addition to architectural enhancements, research on integrating external modal information primarily focuses on leveraging pre-trained models and selective feature extraction to incorporate additional semantic information:

- **RepsNet** (Tanwani, Barral, & Freedman, 2022): Integrates VQA-based answer information with image features, demonstrating how external semantic data can boost content accuracy.

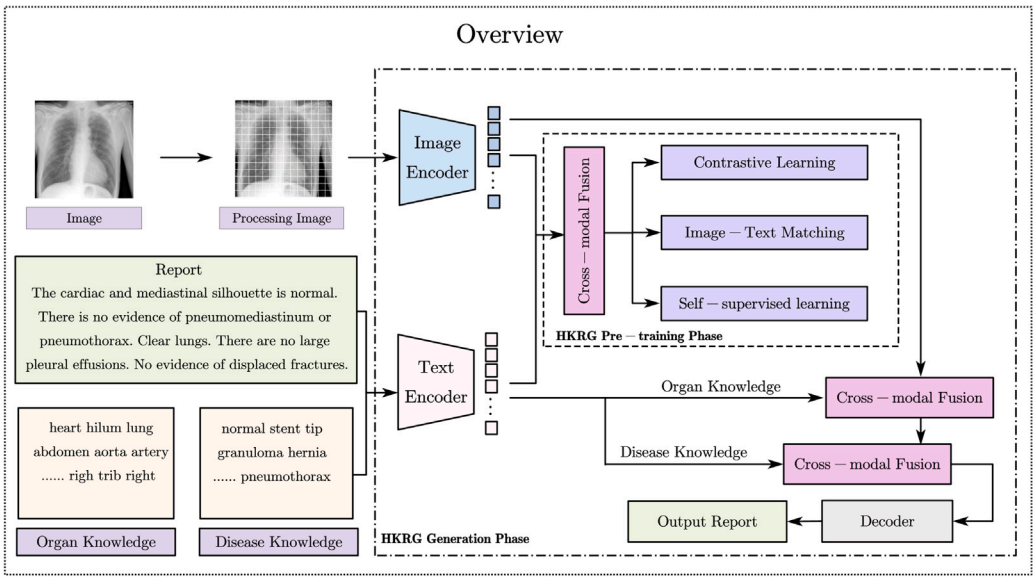


Fig. 2. Workflow of the proposed method. During the training process, our proposed method unfolds in two sequential stages: (1) pre-training phase based on radiology images, reports, and medical knowledge; (2) radiology report generation phase based on hierarchical knowledge integration.

- **RGRG** (Tanida, Müller, Kaissis, & Rueckert, 2023): Focuses on abnormal regions for report generation to reduce redundancy, selectively attending to key areas and improving efficiency and specificity.

Despite the progress in leveraging multi-modal data, existing studies often overlook the fine-grained relationships between anatomical structures and textual descriptions. In particular, current approaches lack the ability to achieve multi-level alignment, encompassing organ-level details, associated pathologies, and their corresponding linguistic representations. Our approach aims to bridge this gap by incorporating hierarchical radiological knowledge to guide report generation, ensuring better alignment between medical images and corresponding textual features.

2.2. Vision-language pre-training

Vision-language pre-training (VLP) models have significantly propelled the capabilities of visual recognition systems, demonstrating profound success across various applications. Typically, VLP models adopt one of two structural approaches: single-stream (Li, Yatskar, Yin, Hsieh, & Chang, 2019) and two-stream architectures (Li et al., 2021). Single-stream methods integrate visual and textual data into a unified model architecture, facilitating direct interaction between the modalities. Conversely, two-stream methods maintain separate pathways for visual and textual data before merging features for final tasks, which can preserve the distinct properties of each modality while still benefiting from joint learning.

In the medical sector, VLP model adaptations have started taking shape, addressing domain-specific challenges:

- **ConVIRT** (Zhang, Jiang, Miura, Manning, & Langlotz, 2022): Pioneered contrastive learning in a medical context, establishing a foundational approach for subsequent models.
- **LoVT** (Chauhan et al., 2020): Focused on enhancing local alignment performance between medical images and corresponding textual descriptions.
- **GLoRIA** (Huang, Shen, Lungren, & Yeung, 2021): Improved local alignment between medical images and text by enhancing regional representations.
- **MedKLIP** (Wu, Zhang, Zhang, Wang, & Xie, 2023): Leveraged disease descriptions to significantly enrich representation learning.

However, these models often struggle to effectively incorporate broad medical knowledge and generalize to unseen data.

Recent advancements in Medical Named-Entity-Recognition (NER) (Chen, Dan, Lu, Chen, & Zhang, 2024) have opened new avenues for extracting key clinical concepts and relationships from radiological reports, which can be structurally formatted to retain essential information. These NER tools provide a valuable resource for enhancing VLP models, offering a structured way to introduce clinical semantics into pre-training, which has been underexplored in existing VLP research.

Although significant progress has been made, existing models still struggle to effectively incorporate broad medical knowledge and generalize to unseen data. They often fail to establish multi-level semantic alignment across visual inputs, domain-specific knowledge, and textual outputs, limiting their capacity to produce clinically meaningful representations. Our approach addresses these challenges by achieving fine-grained alignment between images and knowledge, as well as between images and text, thereby enhancing the semantic depth of vision-language pre-training in the medical domain.

3. Method

In this section, we first provide an overview of our proposed framework, HKRG, which is composed of two main stages: vision-language pre-training and radiology report generation. We begin with an overview of the framework and then provide detailed descriptions of its key components.

3.1. Overview of our proposed method

In radiology report generation tasks, our objective is to generate a descriptive radiology report $R = \{y_1, y_2, \ldots, y_n\}$ based on a radiology image encoded as I. However, the imbalance between normal and abnormal regions in radiology images complicates the multi-granularity alignment between images and reports. To address this, we propose HKRG, a hierarchical knowledge integration framework that guides multi-level image representation learning using radiological knowledge and aligns these representations with report features, as illustrated in Fig. 2.

Our HKRG framework leverages the Swin-Transformer (Liu et al., 2021a) as the image encoder and the KEBERT model (Zhang, Wu, Zhang, Xie, & Wang, 2023), pre-trained on medical knowledge, as the

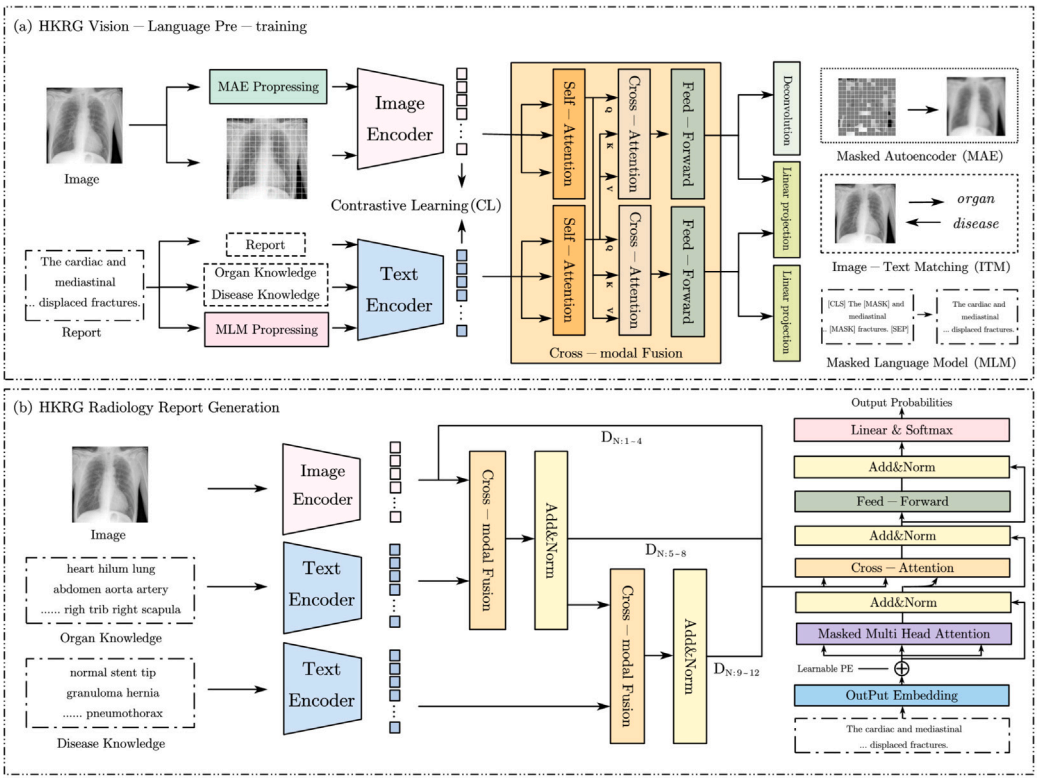


Fig. 3. Illustration of our hierarchical knowledge integration framework (HKRG) with two main stages: (a) vision-language pre-training, and (b) the radiology report generation. Specifically, during the pre-training, the HKRG framework is dedicated to the content of images, radiological knowledge, and the alignment between the two. This hierarchically structured knowledge is effectively integrated and infused into the model through a well-trained fusion mechanism, generating accurate and informative radiology reports.

text encoder. We implement a vision-language pre-training strategy based on multi-task learning to align multi-modal features during the HKRG model's generation phase. Initially, contrastive learning between image features and text features reduces the gap between these modalities, aligning features at the overall image and report levels. This task, aided by organ and disease labels extracted from reports, strengthens the link between visual information and textual descriptions. For fine-grained alignment, we use a multi-modal fusion mechanism where knowledge supplements image features to optimize image–text matching. We also introduce Masked Autoencoder and Masked Language Model tasks on images and text to enhance feature extraction and generation tasks.

During the radiology report generation phase, we first process the radiology image I using an image encoder, segmenting it into multiple blocks and extracting the image features f_I. Concurrently, radiological knowledge related to organs (K_o) and diseases (K_d) is encoded into corresponding knowledge features f_{Ko} and f_{Kd}. Through a cross-modal fusion mechanism, we integrate the image and knowledge features to form multi-modal features that contain both image information and knowledge. These features are then hierarchically fed into a transformer decoder, simulating the diagnostic process of a radiologist: starting with an observation of the overall image, analyzing the image based on organ labels, assigning disease labels to each organ, and ultimately generating a detailed radiology report.

3.2. Radiology report generation

In this section, we describe the key components involved in the radiology report generation process, focusing on hierarchical knowledge integration and cross-modal fusion. As shown in Fig. 3. Hierarchical knowledge integration aims to link radiological images with corresponding report details, while cross-modal fusion enhances the alignment between visual and textual modalities to improve the overall quality of generated reports. We begin by detailing the hierarchical knowledge integration process, followed by an explanation of the cross-modal fusion mechanism.

3.2.1. Hierarchical knowledge integration

We propose a novel approach that links radiological images with the corresponding details in radiology reports through hierarchical knowledge integration. This method enhances the connection between visual features and textual descriptions by introducing radiological knowledge. By progressively integrating organ-level and disease-level knowledge, it improves the performance of abnormality localization and clinical relevance. Specifically, we integrate the radiology image feature f_I, organ knowledge feature f_{Ko}, and disease knowledge feature f_{Kd} using a cross-modal fusion mechanism (CMF). This multi-stage integration ensures that salient anatomical structures are accurately identified and their corresponding abnormalities are more precisely described in the report.

First, the image feature and organ knowledge feature are fused to generate the image-organ representation $f_{(I,Ko)}$, which is then fused with the disease knowledge feature to produce the final multi-modal representation $f_{(I,Ko,Kd)}$:

$$f_{(I,Ko)} = CMF(f_I, f_{Ko}), \tag{1}$$

$$f_{(I,Ko,Kd)} = CMF(f_{(I,Ko)}, f_{Kd}), \tag{2}$$

where $CMF(\cdot)$ represents the cross-modal fusion mechanism that integrates features from different modalities to produce more informative representations.

These fused features are hierarchically fed into the Transformer decoder layer Dec to gradually incorporate radiological knowledge and enhance the consistency between the image and the generated report:

$$Dec\left(f, y_i\right) = \left(f_{(I,Ko,Kd)}, \left(f_{(I,Ko)}, \left(f_I, y_i\right)\right)\right), \tag{3}$$

where f_I represents the feature set at various levels of multi-modal characteristics, and y_i denotes the current prediction target in the report. This hierarchical decoding process allows the model to incrementally refine the report content, making the final output more clinically accurate and contextually relevant.

3.2.2. Cross-modal fusion

We adopt a dual-stream cross attention mechanism in the cross modal fusion module to integrate images and text semantic features. Specifically, in the self-attention layer S_{att}, the feature representations of the single modalities f_I and f_T interact through the internal self-attention mechanism, enhancing the internal representations of each modality:

$$f'_I = S_{att}\left(f_I, f_I, f_I\right), \quad f'_T = S_{att}\left(f_T, f_T, f_T\right), \tag{4}$$

where $Q = K = V = f_I$ is used for the self-attention of the visual modality, and $Q = K = V = f_T$ is used for the self-attention of the textual modality. In the cross-attention layer C_{att}, the process of cross-modal interaction is implemented, where information from different modalities is integrated into their respective representations.

$$f_{Ic} = C_{att}\left(f'_I, f'_T, f'_T\right), \quad f_{Tc} = C_{att}\left(f'_T, f'_I, f'_I\right), \tag{5}$$

where f'_I serves as the query for integrating information from the language modality f'_T, producing the cross-attended visual feature f_{Ic}. Similarly, f'_T serves as the query for integrating information from the visual modality f'_I, producing the cross-attended language feature f_{Tc}. Finally, f_{Ic} and f_{Tc} are fed into the feed forward layer $FFN(\cdot)$ to generate the multi-modal representation of the visual modality f_{FI} and the multi-modal representation of the language modality f_{FT}.

In the radiology report generation phase, we stack the image and text features to generate the final multi-modal representation for report generation.

$$f_{(I,T)} = Concat(f_{FI}, f_{FT}), \tag{6}$$

where $Concat$ denotes the concatenation of feature vectors from different modalities.

$$f_{FI} = \mathrm{FFN}(f_{Ic}), \quad f_{FT} = \mathrm{FFN}(f_{Ic}), \tag{7}$$

In the radiology report generation phase, we stack the image and knowledge features to generate the final multi-modal representation for report generation.

$$f_{(I,K)} = Concat\left(Z_I, Z_K\right), \tag{8}$$

where $Concat$ denotes the concatenation of feature vectors from different modalities, and Z_K represents the multi-modal representation of knowledge obtained through the CMF module.

3.3. Vision-language pre-training

We have employed multiple proxy tasks to pre-train image and text encoders, aimed at deeply exploring the intrinsic relationships and salient features between medical images, associated knowledge, and textual reports. To this end, we employ understanding-based contrastive learning (CL) (Radford et al., 2021) sharpens the model's ability to distinguish relevant from irrelevant image–text-knowledge associations, ensuring fine-grained semantic alignment. Image–text matching (ITM) (Karpathy & Fei-Fei, 2015) further enhances global consistency by training the model to accurately pair visual features with their corresponding textual descriptions. Inspired by SLIP (Mu, Kirillov, Wagner, & Xie, 2022) and M3AE (Chen et al., 2022a), we employ generative masked autoencoders (MAE) (He et al., 2022) to emphasize salient visual details through masked region reconstruction, thereby reinforcing the focus on key image–text correspondences. Simultaneously, Masked Language Modeling (MLM) (Devlin, 2018; Zhang, Jin, Liu, & Xue, 2024) deepens the understanding of contextual relationships and enhances the alignment between medical prior knowledge and visual features. The pre-training objectives are jointly optimized by minimizing the overall loss function, defined as:

$$L_{total} = L_{CL} + L_{ITM} + L_{MAE} + L_{MLM}, \tag{9}$$

where L_{total} represents the overall loss function during the vision and language pre-training phase of the HKRG framework.

3.3.1. Contrastive learning

The contrastive learning task aims to capture the intrinsic connections and external features between radiology images and reports, achieving feature alignment. We map the image features f_I and text features f_T into a shared embedding space z, minimizing the distance between similar image–text pairs while maximizing the distance between dissimilar pairs, thereby effectively aligning the overall features.

$$L_{CL} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{\exp(\mathrm{sim}(\mathbf{I}_i, \mathbf{T}_i)/\tau)}{\sum_{j=1}^{N}\exp(\mathrm{sim}(\mathbf{I}_i, \mathbf{T}_j)/\tau)} + \log\frac{\exp(\mathrm{sim}(\mathbf{T}_i, \mathbf{I}_i)/\tau)}{\sum_{j=1}^{N}\exp(\mathrm{sim}(\mathbf{T}_j, \mathbf{I}_i)/\tau)}\right], \tag{10}$$

where $\mathrm{sim}(\mathbf{I}_i, \mathbf{T}_i)$ represents the similarity between the image $\mathbf{I}_i$ and the corresponding text $\mathbf{T}_i$ in the shared embedding space, τ is a temperature parameter, and N is the total number of image–text pairs in the batch.

3.3.2. Image–text matching

The image–text matching task aims to learn multi-modal representations that capture fine-grained alignments between image and knowledge. We combine images with radiology knowledge (such as organ labels and disease labels) through a cross-modal fusion mechanism, generating the image feature I_c and knowledge feature K_c. These features are then processed through a linear layer to determine whether the image matches the corresponding knowledge. The task performance is evaluated using the loss function L_{ITM}, as described by the following formula:

$$L_{ITM} = \mathbb{E}_{(I_c,K_c)\sim D}\left[H\left(\mathbf{y}^{\mathrm{itm}}, \mathbf{p}^{\mathrm{itm}}(I_c, K_c)\right)\right], \tag{11}$$

where $\mathbb{E}_{(I_c,K_c)\sim D}$ represents the expectation over the distribution D of image feature I_c and text feature K_c. $H(\cdot,\cdot)$ denotes the cross-entropy loss function. $\mathbf{y}^{\mathrm{itm}}$ is the actual matching label, and $\mathbf{p}^{\mathrm{itm}}(I_c, K_c)$ is the model's predicted matching probability for image I_c and text K_c.

3.3.3. Masked autoencoder

The MAE task captures the intrinsic representation of data by randomly masking 70% of the image patches and requiring the model to reconstruct them. Unlike traditional MAE methods, HKRG shares the image encoder during the pre-training phase and introduces a cross-modal mechanism during training to integrate textual information. By incorporating report data, it effectively compensates for the lack of contextual information in the images. To reconstruct the images, the model uses deconvolution layers as the decoder and L_{MAE} as the loss function:

$$L_{MAE} = \mathbb{E}_{(I,M)\sim D}\left[\frac{1}{|M|}\sum_{i\in M}\left(I_i - \hat{I}_i\right)^2\right], \tag{12}$$

where I_i and $\hat{I}_i$ represent the pixel values of the ith patch in the original and reconstructed images, respectively. M denotes the set of indices corresponding to the masked patches in the image. $|M|$ is the number of masked patches.

3.3.4. Masked language model

The Masked Language Model task aims to learn the contextual relationships between different parts of the input data by predicting masked words or image regions. In our framework, the MLM task enhances the representation ability of the text encoder by learning the relationships between contexts, while providing feedback to the other pre-training tasks, strengthening the consistency between image features and text knowledge. We randomly mask 15% of the input tokens and replace them with the [MASK] token. The MLM is trained by minimizing the following cross-entropy loss function L_{MLM}:

$$L_{MLM} = \mathbb{E}_{(I,\hat{T})\sim D}\left[H\left(\mathbf{y}_{\text{msk}}, p_{\text{msk}}(I,\hat{T})\right)\right], \tag{13}$$

where $\hat{T}$ represents the masked text, with $p_{\text{msk}}(I,\hat{T})$ indicating the model's predicted probabilities for the masked tokens, $\mathbf{y}_{\text{msk}}$ is a one-hot vocabulary distribution where the true token has a probability of 1.

4. Experimental results

In this section, we first introduce two public datasets as well as some of the experimental setup, and the evaluation metrics. Then, we assess the effectiveness and advantages of each key component through ablation experiments and qualitative analysis.

4.1. Dataset

IU-Xray (Demner-Fushman, Kohli, Rosenman, Shooshan, Rodriguez, Antani, Thoma, & McDonald, 2016) from Indiana University is used to evaluate the performance of radiology report generation methods. During data preparation, following previous research (Chen et al., 2020; Huang et al., 2023), we first excluded entries without the "Findings" section, ultimately retaining 6471 images and 3336 reports. Then, we randomly split the dataset into training (70%), validation (10%), and test (20%) sets, ensuring no overlap of patients between these sets. Finally, we preprocessed the reports by tokenizing, converting to lowercase, and removing non-alphabetic characters.

MIMIC-CXR (Johnson et al., 2019) provided by the Beth Israel Deaconess Medical Center is a recently released large-scale dataset. We use the official splits to report the results. The dataset is divided into 368,960 training instances, 2991 validation instances, and 5159 testing instances, ensuring no overlap of patients between the training, validation, and test sets. Finally, we preprocess the reports by tokenizing, converting all words to lowercase, and removing special characters.

For the pre-training experimental data, we first constructed a medical knowledge base for each dataset. Relevant medical terms were automatically extracted from radiology reports to ensure comprehensive and accurate coverage. These extracted terms were then manually verified by professional radiologists to guarantee the accuracy of the knowledge base, providing high-quality medical background knowledge for the model. Finally, the training set and the verified medical knowledge base were used as the pre-training data for HKRG.

4.2. Implementation details

Our method was developed under the Pytorch 1.12.1 framework, utilizing two 24 GB NVIDIA GeForce RTX 4090 GPUs. The multi-task self-supervised learning approach involves Masked Autoencoding (MAE) for images, with a masking ratio of 70%, and Masked Language Modeling (MLM) for texts, applying a masking ratio of 15%. The multi-modal fusion module employs a 6-layer Transformer with 768 hidden dimensions and 8 attention heads to integrate features from both modalities effectively. The training process for report generation utilizes a 12-layer Transformer decoder that incorporates image features alongside anatomical and pathological knowledge in a segmented phase-wise manner. During the pre-training phase, the model was trained for 500 epochs, and the subsequent training phase consisted of 300 epochs. The entire architecture is trained with the Adam optimizer, using an initial learning rate of 1×10^{-5}, a batch size of 16, and an input size of 256×256. The learning rate schedule includes a 10% warm-up period, followed by a linear decay schedule.

4.3. Metrics

We evaluate the performance of radiology report generation using several widely adopted natural language generation (NLG) metrics. BLEU (Papineni, Roukos, Ward, & Zhu, 2002) measures the lexical overlap between the generated report and the reference report, METEOR (Banerjee & Lavie, 2005) accounts for factors such as stemming and synonyms to assess semantic similarity, ROUGE-L (Lin, 2004) focuses on the global structural similarity between the two reports, and Clinical Efficacy (CE) gauges the presence of significant clinical observations that reflect the diagnostic accuracy of the generated reports.

To evaluate the statistical significance of differences between models, we conducted paired t-tests for each evaluation metric. A p-value of less than 0.05 was considered significant. This threshold is based on conventional standards in machine learning research, aiming to balance Type I and Type II errors and facilitate comparisons with existing studies.

4.4. Comparison with state-of-the-art

In the comparative experiments, our HKRG model was benchmarked against the latest advanced models on the IU-Xray and MIMIC-CXR datasets. As shown in Table 1, our HKRG model outperforms the state-of-the-art methods across all natural language generation metrics on the IU-Xray dataset. Specifically, the Bleu1 score of HKRG improved from 0.536 to 0.538 (an increase of 0.37%), the Bleu4 score from 0.206 to 0.213 (an increase of 3.4%), and the Meteor score by approximately 0.83%. These results validate the effectiveness of injecting multi-level medical knowledge during the decoding phase, enabling the model to perform multi-level knowledge learning based on the sequence and location of images, organs, and pathologies. Compared to models using similar pre-training techniques (Yan & Pei, 2022; Zhan et al., 2024), our model has begun to deeply understand and categorize images and medical texts during the pre-training phase, significantly enhancing its performance in the task of generating medical reports. Furthermore, compared to methods specifically designed for radiological report generation (Chen et al., 2022b, 2020; Huang et al., 2023; Liu et al., 2021b; Wang et al., 2023; You et al., 2021), our cross-modal fusion mechanism more effectively learns the relationships between text and images through fine-grained alignment of image features and knowledge levels.

Our HKRG model also performed excellently on the MIMIC-CXR dataset, as shown in Tables 1 and 2. Although our model underperforms the COMG model in terms of Bleu2 and Bleu3 metrics and slightly lags behind the MedM2G model in precision and F1 scores, it still surpasses the most advanced methods in other metrics. This performance variation is attributed to the reinforcement learning mechanism of the COMG model, which excels particularly in the large-scale MIMIC-CXR dataset by learning optimal strategies that enhance decision-making in clinical observations. However, this mechanism might not show the same effectiveness in the smaller IU-Xray dataset due to limited data. The main advantage of the MedM2G model lies in its cross-modal generative capabilities, allowing it not only to learn features of X-ray images but also to capture feature information across different modalities of the same image. This capability relies heavily on large-scale data for optimal learning, which may limit its effectiveness in smaller datasets. These comparisons highlight our model's excellent generalization ability across multiple environments, reinforcing its potential for widespread clinical application.

4.5. Ablation studies

To verify the effectiveness of the key components of the HKRG framework, we conducted detailed ablation studies using the IU-Xray dataset.

Table 1
Performance comparison of HKRG method and existing methods on IU-Xray (upper part) and MIMIC-CXR (lower part) test datasets. The best values are highlighted in bold.

Data	Method	NLG metrics					
		Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge L
IU-Xray	R2Gen (Chen et al., 2020)	0.470	0.304	0.219	0.165	0.187	0.371
	R2GenCMN (Chen et al., 2022b)	0.475	0.309	0.222	0.170	0.191	0.375
	PPKED (Liu et al., 2021b)	0.483	0.315	0.224	0.168	0.190	0.376
	AlignTrans (You et al., 2021)	0.484	0.313	0.225	0.173	0.204	0.379
	Clinical-BERT (Yan & Pei, 2022)	0.495	0.330	0.231	0.170	–	0.376
	METransformer (Wang, Liu, Wang, & Zhou, 2023)	0.483	0.322	0.228	0.172	–	0.380
	Kiut (Huang et al., 2023)	0.525	0.360	0.251	0.185	0.242	0.409
	COMG (Gu, Liu, Li, & Cai, 2024)	0.536	0.378	0.275	0.206	–	0.383
	MedM2G (Zhan et al., 2024)	0.533	0.369	0.278	0.212	–	0.416
	HKRG (ours)	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**
MIMIC-CXR	R2Gen (Chen et al., 2020)	0.353	0.218	0.145	0.103	0.142	0.277
	R2GenCMN (Chen et al., 2022b)	0.353	0.218	0.148	0.106	0.142	0.278
	PPKED (Liu et al., 2021b)	0.360	0.224	0.149	0.106	0.149	0.284
	AlignTrans (You et al., 2021)	0.378	0.235	0.156	0.112	0.158	0.283
	Clinical-BERT (Yan & Pei, 2022)	0.383	0.230	0.151	0.106	–	0.275
	METransformer (Wang et al., 2023)	0.386	0.250	0.169	0.124	–	0.291
	Kiut (Huang et al., 2023)	0.393	0.243	0.159	0.113	0.160	0.285
	COMG (Gu et al., 2024)	0.363	0.235	0.167	0.124	–	0.290
	MedM2G (Zhan et al., 2024)	0.412	**0.260**	**0.179**	0.142	–	0.309
	HKRG (ours)	**0.417**	0.253	0.172	**0.143**	**0.167**	**0.310**

Table 2
Comparison of clinical efficacy metrics for the MIMIC-CXR dataset. The best values are highlighted in bold.

Method	CE Metric		
	Precision	Recall	F1
R2Gen	0.333	0.273	0.276
R2GenCMN	0.334	0.275	0.278
METransformer	0.364	0.309	0.311
Kiut	0.371	0.318	0.321
COMG	**0.424**	0.291	**0.345**
HKRG (ours)	0.384	**0.324**	0.339

4.5.1. Pre-training of key components

To evaluate the contribution of each key component during the pre-training phase to the model's performance, we conducted an ablation experiment on the IU-Xray dataset, with the results shown in Table 3. The experiment systematically analyzed the impact of each component on natural language generation (NLG) metrics by gradually removing or adding different pre-training loss functions (L_{CL}, L_{ITM}, L_{MAE}, L_{MLM}).

The baseline configuration did not use any pre-training loss functions and only employed the generation phase strategies HKI and CMF, resulting in Bleu1 and Bleu4 scores of 0.514 and 0.170, respectively. After introducing L_{CL} during the pre-training phase, the Bleu1 and Bleu4 scores improved to 0.518 and 0.176, benefiting from the cross-modal alignment effect. Building on this, adding the L_{ITM} task led to a significant further improvement in Bleu1 and Bleu4 scores. However, when only the ITM task was used for pre-training, the metrics dropped compared to the CL task. This is because the ITM task further enhances the alignment between image and text, particularly in the medical domain, where matching images with medical knowledge helps improve semantic consistency between images and texts, leading to more accurate image descriptions.

When we pre-trained the image encoder and text encoder using the self-supervised tasks MAE and MLM, the performance improvement was relatively small compared to the baseline model across all metrics, indicating that separately pre-training the image and text encoders did not significantly improve the generation phase. However, after adding the CL and ITM tasks, the performance increased significantly. Compared to the baseline method, BLEU-1 improved from 0.514 to 0.538, a 4.68% increase, and BLEU-4 improved from 0.170 to 0.196, a 15.29% increase. The ablation experiment during the pre-training phase shows that each pre-training component contributes positively to improving the model's NLG performance, especially the combination of contrastive loss and image–text matching loss, which significantly enhances the generation capability. The comprehensive integration of all components achieves the best performance, proving the effectiveness of the integrated pre-training strategy and the synergistic effect among the components.

4.5.2. Report generation of key components

As shown in Table 3, We conducted a comprehensive evaluation of the impact of Hierarchical Knowledge Injection (HKI) and Cross-Modal Fusion (CMF) mechanisms on model performance. Without the integration of HKI and CMF, the model showed a decrease in performance, with a Bleu1 score of 0.503 and a Bleu4 score of 0.182. Upon incorporating HKI, there was a noticeable performance improvement: the Bleu1 score increased to 0.517 and the Bleu4 score rose to 0.217. This enhancement was due to the integration of hierarchical knowledge, which effectively facilitated multi-level alignment between image features and text reports. Further use of the complete HKRG model elevated the Bleu1 score to 0.538 and the Bleu4 score to 0.213. This improvement was attributed to the synergistic effect of the cross-modal fusion mechanism and HKI, enabling the model to combine knowledge acquired during pre-training with the internal and external features of images. This guided the representation learning of images, generating accurate radiological reports.

4.5.3. Hierarchical knowledge analysis

To further verify the effectiveness of the HKI module in our method. By incrementally introducing key components at different levels, we designed four different experimental settings. Baseline: Our baseline method employs a traditional multimodal Transformer framework. Setting1: In the baseline method, we added organ knowledge features $f_{(I,Ko)}$ and CMF mechanism. Setting2: We added disease knowledge features $f_{(I,Kd)}$ and cross-modal fusion mechanism to the baseline method. HKMR (ours): Our proposed complete method, incorporates both organ and disease knowledge features and the cross-modal fusion mechanism into the baseline.

As shown in Table 4, we conducted a paired t-test to statistically analyze each variable compared to the baseline method. The p-values in the table are all less than 0.05, indicating that the methods incorporating key components significantly outperform the baseline in radiology

Table 3
Key component ablation analysis in the IU-Xray dataset. A statistical significance study using paired t-test (p-value < 0.05). The best values are highlighted in bold.

Stage	Key components						NLG metrics						
	Pre-training				Generation		Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge L	P-value
	L_{CL}	L_{ITM}	L_{MAE}	L_{MLM}	HKI	CMF							
Pre-training	×	×	×	×	✓	✓	0.514	0.356	0.238	0.170	0.193	0.371	–
	✓	×	×	×	✓	✓	0.518	0.360	0.240	0.176	0.204	0.388	5.8e−3
	×	✓	×	×	✓	✓	0.516	0.358	0.238	0.172	0.196	0.378	5.4e−3
	✓	✓	×	×	✓	✓	0.520	0.364	0.252	0.182	0.216	0.393	6.2e−3
	×	×	✓	✓	✓	✓	0.514	0.357	0.239	0.174	0.192	0.375	4.1e−3
	✓	✓	✓	×	✓	✓	0.531	0.372	0.264	0.191	0.231	0.394	7.9e−3
	✓	✓	×	✓	✓	✓	0.526	0.366	0.267	0.188	0.226	0.396	8.2e−3
Report Generation	✓	✓	✓	✓	×	×	0.503	0.331	0.228	0.182	0.190	0.376	7.7e−4
	✓	✓	✓	✓	✓	×	0.517	0.351	0.252	0.204	0.217	0.398	5.2e−3
HKMR (ours)	✓	✓	✓	✓	✓	✓	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**	3.6e−4

Table 4
Ablation analysis of the key component hierarchical knowledge integration on the IU-Xray dataset.

Settings	Key components			CMF	NLG metrics						p-value
	f_I	$f_{(I,Ko)}$	$f_{(I,Kd)}$		Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge L	
Baseline	✓	×	×	×	0.503	0.331	0.228	0.182	0.190	0.376	4.1e−3
Setting1	✓	✓	×	✓	0.519	0.363	0.253	0.195	0.216	0.381	5.6e−3
Setting2	✓	×	✓	✓	0.524	0.368	0.257	0.202	0.223	0.383	7.8e−3
HKMR (ours)	✓	✓	✓	✓	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**	4.7e−4

report generation.

Compared to the baseline model that only uses basic image features, the settings that progressively introduce anatomical and pathological knowledge show improvements across all metrics. We introduced organ knowledge into the baseline model in Setting 1 to mitigate the impact of significant variations in organ shape, size, and position across patients (e.g., differences in liver size and position, changes in kidney shape, or variations in the heart's position relative to the chest) that may affect the accuracy of feature extraction and model generalization. After incorporating organ knowledge features ($f_{(I,Ko)}$) and the CMF mechanism, a 1% to 9% improvement in various metrics was achieved compared to the baseline, demonstrating enhanced adaptability to structural differences.

Furthermore, variations in image quality, such as noise and blurriness due to different imaging devices or scanning parameters, can adversely affect model performance by obscuring critical features necessary for accurate report generation. Setting2, which guides image feature learning through disease label knowledge ($f_{(I,Kd)}$) and includes the cross-modal fusion mechanism, led to an increase of 0.21 in Bleu1 and 0.20 in Bleu4. This indicates improved robustness against such quality variations, as the disease label knowledge helps the model focus on relevant pathological regions despite image degradation.

Finally, the HKMR model, which combines complete hierarchical knowledge integration and the cross-modal fusion module, achieved the best performance across all metrics, with Bleu1 and Rouge L reaching 0.538 and 0.418, respectively, significantly outperforming other configurations. This is mainly attributed to the integration of organ knowledge and disease knowledge. The model first extracts preliminary features from radiological images using organ knowledge, and then further makes judgments about the patient based on disease knowledge. The two sources of knowledge are interconnected during the learning process, effectively guiding the model to learn the image features of abnormal regions.

4.6. Comparative analysis of pretraining methods

This section compares the performance of our proposed HKRG method with existing pretraining methods on two benchmark datasets: IU-Xray and MIMIC-CXR. As shown in Table 5, the experimental results demonstrate consistent improvements in Natural Language Generation (NLG) metrics, validating the effectiveness of our hierarchical knowledge integration strategy.

On the IU-Xray dataset, our method outperforms existing methods across all NLG metrics, particularly achieving significant improvements in Bleu4 (0.213) and Meteor (0.244). Compared to MedKLIP (Wu et al., 2023), our method shows an 8.1% improvement in Bleu4 and a 15.1% improvement in Meteor. These significant gains are attributed to the fine-grained alignment of visual features and radiological knowledge during pretraining, which enhances the precision of the generated reports. On the MIMIC-CXR dataset, our method surpasses others in key metrics such as Bleu4 (0.143) and Rouge L (0.310), and remains competitive across other metrics. Notably, the improvement in Rouge L indicates that our model is able to generate reports that are structurally more consistent with the reference reports.

The performance improvement of the HKRG pretraining method, compared to other models, stems from its hierarchical knowledge integration strategy specifically designed for the generation phase of HKRG. Unlike other models that treat the entire text report as a single pretraining dataset, the HKRG method aligns organ knowledge, disease knowledge, and radiology reports in a hierarchical manner, effectively integrating these forms of knowledge into downstream tasks, thereby significantly enhancing the model's performance.

4.7. Comparison of medical knowledge extraction methods

In this experiment, we compared three medical knowledge extraction methods: BioBERT, KEBERT, and our method. As shown in Table 6, our HKRG method outperforms both BioBERT and KEBERT across all evaluation metrics, especially with significant improvements in Bleu-4 and Rouge L, highlighting its advantage in medical knowledge extraction. While BioBERT and KEBERT have pre-training foundations in the medical domain, they primarily focus on extracting semantic information through word embedding models and fail to fully incorporate prior medical knowledge. This limitation affects their ability to handle medical terms and context effectively.

4.8. Hyperparameter analysis

In this subsection, we will delve into the analysis of hyperparameters to study how different settings affect the performance of our model on the IU-Xray dataset for the task of generating radiological reports. We will examine several key hyperparameters, including the number

Table 5
Performance comparison of HKRG pre-training method and existing pre-training methods on IU-Xray (upper part) and MIMIC-CXR (lower part) test datasets. The best values are highlighted in bold.

Data	Method	NLG metrics					
		Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge L
IU-Xray	GLoRIA (Huang et al., 2021)	0.516	0.362	0.249	0.183	0.197	0.386
	MedBLIP (Chen & Hong, 2024)	0.519	0.367	0.257	0.201	0.203	0.402
	MedKLIP (Wu et al., 2023)	0.521	0.368	0.262	0.197	0.212	0.398
	Ours	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**
MIMIC-CXR	GLoRIA (Huang et al., 2021)	0.398	0.233	0.155	0.131	0.147	0.287
	MedBLIP (Chen & Hong, 2024)	0.403	0.243	0.161	0.138	0.156	0.303
	MedKLIP (Wu et al., 2023)	0.405	0.241	0.163	0.128	0.153	0.297
	Ours	**0.417**	**0.253**	**0.172**	**0.143**	**0.167**	**0.310**

Table 6
Comparison of medical knowledge extraction methods.

Method	NLG metrics					
	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge L
BioBERT (Lee et al., 2020)	0.524	0.368	0.254	0.192	0.228	0.392
KEBERT (Zhang et al., 2023)	0.527	0.369	0.256	0.197	0.231	0.397
ours	0.538	0.380	0.278	0.213	0.244	0.418

Table 7
Hyperparameter analysis of decoder layer numbers in the IU-Xray dataset.

Dec_{Nm}	NLG metrics						p-value
	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge L	
6	0.512	0.351	0.259	0.174	0.201	0.388	–
9	0.524	0.362	0.261	0.189	0.224	0.394	3.8e−3
12	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**	2.1e−4
16	0.531	0.374	0.269	0.202	0.237	0.405	1.5e−3

Table 8
Hyperparameter analysis of learning rate in the IU-Xray dataset.

Learning rate	NLG metrics						p-value
	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	RougeL	
1e−4	0.536	0.376	0.277	0.213	0.241	0.416	–
1e-5	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**	3.5e−3
8e−6	0.532	0.374	0.274	0.209	0.242	0.413	4.2e−3

of decoder layers, learning rate, optimizer type, and regularization strategies. By systematically varying each parameter, we will observe its impact on model performance.

4.8.1. Hyperparameter analysis of decoder layer numbers

Table 7 shows the impact of different decoder layer numbers (Dec_{Nm}) on the NLG metrics of the model. As the number of layers increases, we observe improvements in most NLG metrics, indicating that a deeper decoder enhances the model's ability to generate more accurate and coherent radiology reports. Specifically, we distribute the image features f_I, multi-modal features combining image and organ knowledge $f_{(I,Ko)}$, and multi-modal features combining image, organ, and disease knowledge $f_{(I,Ko,Kd)}$ evenly across the decoder layers. When $Dec_{Nm} = 12$, the model achieves the best performance across all metrics, with Bleu1, Bleu4, Meteor, and Rouge L scores of 0.538, 0.213, 0.244, and 0.418, respectively. This suggests that the 12-layer decoder strikes an optimal balance between model complexity and performance. However, when the number of layers is further increased to 16, there is a slight decline in performance, with reductions in Bleu1, Bleu4, Meteor, and Rouge L scores. This indicates that while a deeper model can capture more complex patterns, it may also increase the difficulty of optimization, leading to a drop in performance.

4.8.2. Hyperparameter analysis of learning rate

In Table 8, we present the impact of different learning rates on the task of generating radiological reports for the IU-Xray dataset. By comparing the results at learning rates set at 1×10^{-4}, 1×10^{-5}, and 8×10^{-6}, we find that the model performs slightly better on all evaluation metrics when the learning rate is 1×10^{-5}. These results suggest that selecting 1×10^{-5} as the initial learning rate can significantly enhance the model's generation quality for this dataset and task.

4.8.3. Hyperparameter analysis of optimizer type

In Table 9, we present a hyperparameter analysis comparing three different optimizers SGD, RMSprop, and Adam on their performance in generating radiological reports for the IU-Xray dataset. The impact of different optimizers on model training and final output is significant. Among them, the Adam optimizer shows the best performance across all evaluation metrics, indicating its greater efficiency in adjusting model parameters within the HKRG framework.

4.8.4. Hyperparameter analysis of regularization strategy

In Table 10, we analyzed the impact of different regularization strategies on the task of generating radiological reports for the IU-Xray dataset. By comparing models without regularization to those with varying degrees of Dropout regularization (p=0.2 and p = 0.5), the results show that applying 20% Dropout regularization improved all natural language generation metrics compared to the model without regularization: Bleu1 increased to 0.538, Bleu2 to 0.380, Bleu3 to 0.278, Bleu4 to 0.213, Meteor to 0.244, and RougeL to 0.418. However, increasing the Dropout rate to 50% resulted in performance improvements over the non-regularized model, but lower than the Dropout (p=0.2) configuration. Specifically, the metrics were Bleu1 at 0.527, Bleu2 at 0.367, Bleu3 at 0.269, Bleu4 at 0.202, Meteor at

Table 9
Hyperparameter analysis of optimizer type in the IU-Xray dataset.

Optimizer type	NLG metrics						p-value
	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	RougeL	
SGD	0.527	0.377	0.274	0.208	0.236	0.404	–
RMSprop	0.534	0.376	0.271	0.210	0.238	0.407	4.2e−3
Adam	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**	2.7e−3

Table 10
Hyperparameter analysis of regularization strategy in the IU-Xray dataset.

Regularization strategy	NLG metrics						p-value
	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	RougeL	
None	0.513	0.359	0.251	0.192	0.213	0.391	–
Dropout (p=0.2)	**0.538**	**0.380**	**0.278**	**0.213**	**0.244**	**0.418**	7.5e−4
Dropout (p=0.5)	0.527	0.367	0.269	0.202	0.234	0.403	2.3e−4

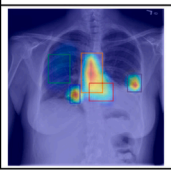
Radiology Image	Image Feature f_I	Ground Truth	HKRG
		Heart size and mediastinal contour are normal. Pulmonary vascularity is normal. The right lung is clear. There is a recurrence moderate – sized left pleural effusion. No pneumothorax. Limited right base stringy density compatible with atelectasis. Dextroscoliosis of the thoracic spine.	Normal heart size and mediastinal contour, pulmonary vascularity is normal. Clear right lung, stringy densities at the right base with atelectasis. Moderate – sized recurrent left pleural effusion is present. No evidence of pneumothorax. Dextroscoliosis of the thoracic spine is noted.

Fig. 4. The visualization of the HKRG, the border colors are aligned with the colors present in the text. Different colors indicate different organs or diseases.

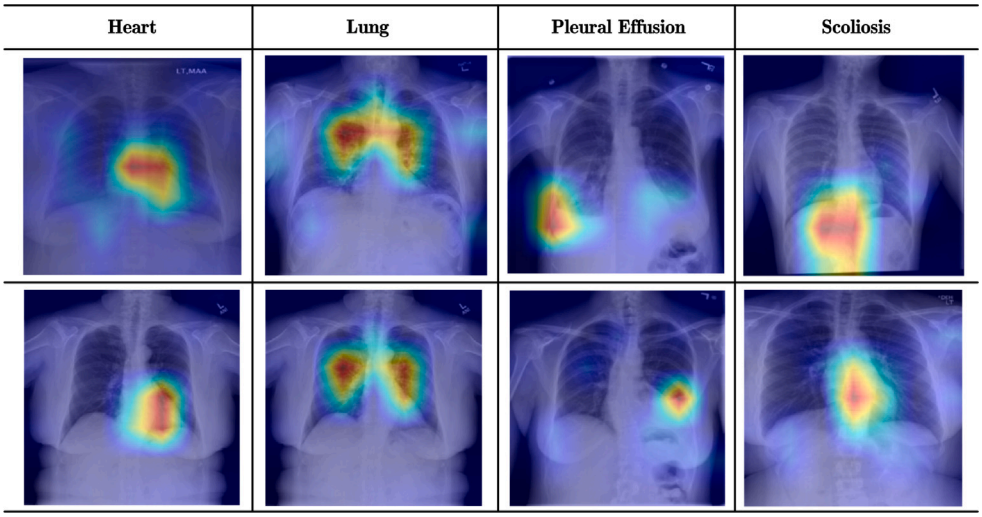


Fig. 5. The visualization of organ and disease attention mechanism.

0.234, and RougeL at 0.403. This indicates that an appropriate level of Dropout (such as 20%) can effectively enhance model quality, while a higher Dropout rate (such as 50%) might lead to performance degradation, suggesting that balance is crucial when choosing regularization strategies.

4.9. Qualitative analysis

To better understand our method visually, we conducted a qualitative analysis of the images, as shown in Fig. 4. HKRG utilizes radiological knowledge to guide the image features to focus on organ structures and related diseases, thereby generating a structured report. Specifically, HKRG first focuses on the organ structures and then shifts attention to the related abnormal areas. For example, the generated report accurately describes the positions of the "right lung" and "left pleural," due to the hierarchical integration of knowledge and interaction with the images. Notably, our report maintains a good balance between descriptions of normal and abnormal areas, even when highlighting features of the normal areas.

In Fig. 5, we provide further visualizations based on the IU-Xray dataset. These visualizations highlight the effectiveness of the knowledge attention mechanism in our model. As depicted in the figure, the model accurately emphasizes the relevant relationships between organs and diseases, demonstrating its ability to focus on critical knowledge components.

5. Conclusion

In this paper, we present a hierarchical knowledge integration framework, HKRG, designed to optimize the radiology report generation process. By integrating multilevel radiological knowledge and employing a specialized pre-training task, HKRG achieves deep alignment between knowledge, images, and textual reports. Our experimental results on two datasets demonstrate that HKRG surpasses current state-of-the-art technologies in generating accurate radiology reports, achieving BLEU4 and ROUGE-L scores of 0.213 and 0.418, respectively.

However, while the model performs well in experimental settings, real-world deployment poses several challenges. First, the quality of data in real-world environments may be inconsistent and noisy, which will require additional data cleaning and fine-tuning of the model. Second, the scalability and real-time performance of the model are critical for practical applications, particularly in healthcare, where efficiently processing large-scale data and providing real-time feedback are essential. In future work, we aim to enhance the integration of fine-grained anatomical knowledge to support the early identification of diseases. Additionally, we plan to advance multimodal fusion technologies to improve the quality and specificity of generated reports. Clinically, enhancing the accuracy of these reports can streamline radiologists' workflows, enabling faster and more reliable diagnoses.

CRediT authorship contribution statement

Bo Wang: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Software, Writing – original draft, Writing – review & editing. **Peihong Teng:** Data curation, Formal analysis, Methodology, Writing – original draft, Project administration, Writing – review & editing. **Hongda Zhang:** Data curation, Formal analysis, Methodology, Software, Writing – review & editing. **Feiyang Yang:** Methodology, Software, Writing – review & editing. **Zeyu Wang:** Methodology, Software, Writing – review & editing. **Xingcheng Yi:** Methodology, Supervision, Methodology, Validation. **Tianyang Zhang:** Supervision, Methodology, Validation. **Chunhui Wang:** Supervision, Methodology, Validation. **Adriano Jose Tavares:** Methodology, Validation, Writing – review & editing. **Hao Xu:** Conceptualization, Methodology, Validation, Resources, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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